

Data Science Capstone Project Report

SpaceX Falcon 9 First Stage Landing Prediction

90

Launches

4

ML Models

83.3%

Accuracy

67%

Success Rate

Presented by

BIKASH

IBM Applied Data Science Capstone

github.com/Bizsubba/IBM-Data-Science-Capstone

Executive Summary

KEY FINDINGS AT A GLANCE

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Business Goal

Predict Falcon 9 first-stage landing to estimate launch cost. SpaceX charges \$62M vs competitors at \$165M — reusability is the key differentiator.



Data Collected

90 Falcon 9 launches via SpaceX REST API + 364 records from Wikipedia scraping. 83 engineered features used for ML modeling.



Best Launch Site

KSC LC-39A achieved the highest success rate at 76.9%. ES-L1, GEO, HEO and SSO orbits all reached 100% success.



Best ML Model

Decision Tree achieved top CV score of 87.5%. All 4 classifiers tied at 83.3% test accuracy — strong generalization.

Success rate improved from 0% (2010) → 90% (2019) — reflecting SpaceX's rapid technological advancement in reusable rocketry

Introduction

WHY DOES FIRST-STAGE RECOVERY MATTER?

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The Business Problem

- SpaceX charges just \$62M per Falcon 9 launch
- Competitors charge up to \$165M per launch
- Savings achieved by reusing the first-stage booster
- Predicting landing success = estimating launch cost
- Valuable for companies bidding against SpaceX

SpaceX
Falcon 9

\$62M

per launch

Competitors

\$165M

per launch

Project Objectives

01

Collect Data

SpaceX REST API + Wikipedia scraping for comprehensive Falcon 9 launch records

02

Explore & Analyse

SQL queries, Matplotlib/Seaborn charts, and Folium interactive maps

03

Build ML Models

Train Logistic Regression, SVM, Decision Tree & KNN classifiers

04

Predict & Deploy

Identify best model to predict landing success for cost estimation

Data Collection Methodology

SPACEX REST API · WIKIPEDIA WEB SCRAPING

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SpaceX REST API

api.spacexdata.com/v4/launches/past

- GET request to SpaceX REST API v4
- 107 total launches retrieved in JSON
- Secondary API calls resolved rocket, payload, launchpad & core IDs
- Filtered to 90 Falcon 9 launches (removed 8 non-Falcon-9)
- Missing PayloadMass imputed with column mean
- 31 missing LandingPad values (early no-attempt launches)

107 → 90 launches

17 features extracted



Wikipedia Web Scraping

en.wikipedia.org — *Falcon 9 & Heavy launches*

- HTTP GET with User-Agent header (bypassed 403 block)
- BeautifulSoup parsed all wikitables HTML tables
- Custom helper functions extracted: date, booster, site, payload, orbit, customer, outcome
- 364 launch records extracted from 2010 to 2024
- Saved as spacex_web_scraped.csv for downstream analysis

364 records

2010 → 2024 coverage

Dual-source strategy ensures comprehensive Falcon 9 launch history with both structured API data and rich historical context

Data Wrangling Methodology

CLEANING · ENCODING · FEATURE ENGINEERING

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01 Filter Falcon 9

Removed Falcon 1 & other variants. Retained 90 Falcon 9 launches from 107 total records.

02 Handle Nulls

Identified missing PayloadMass values. Imputed with column mean to preserve dataset size.

03 Create Class Label

Binary target: 1=success (True ASDS/RTLS/Ocean), 0=failure. Result: 60 successes, 30 failures.

04 One-Hot Encoding

Encoded Orbit, LaunchSite, Serial, GridFins, Reused, Legs → 83 total feature columns.

05 Standardization

Applied StandardScaler across all 83 features to normalize scale before ML training.

06 Train/Test Split

80/20 split (random_state=2): 72 training · 18 test samples for unbiased evaluation.

EDA & Visual Analytics Methodology

FOUR COMPLEMENTARY ANALYTICAL APPROACHES

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EDA with Visualization

Matplotlib · Seaborn

- › Scatter: Flight Number vs Launch Site
- › Scatter: Payload Mass vs Orbit type
- › Bar: Success rate per orbit type
- › Line: Annual success rate trend 2010–2020



EDA with SQL

SQLite · Pandas read_sql

- › 10 SQL queries on SPACEXTBL table
- › Aggregated payload, sites, outcomes
- › Date-range filtered outcome ranking
- › Subquery for maximum payload booster



Folium Interactive Maps

Folium · MarkerCluster

- › Mapped all 4 US launch sites on globe
- › Green/red markers: success vs failure
- › Distance to coastline: 0.93 km (CCAFS)
- › Proximity lines to highway & city



Plotly Dash Dashboard

Plotly Express · Dash Callbacks

- › Dropdown filter by launch site
- › Pie chart: success distribution by site
- › Scatter: payload vs outcome by booster
- › RangeSlider: payload mass 0–10,000 kg

Multi-approach analysis reveals different data dimensions — together providing 360° insight into Falcon 9 launch patterns

Predictive Analysis Methodology

FOUR CLASSIFIERS WITH GRIDSEARCHCV HYPERPARAMETER TUNING

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Logistic Regression

GridSearchCV · 10-fold cross-validation

Parameters:

C: [0.01, 0.1, 1]

penalty: l2 · solver: lbfgs

CV

84.64%

Test

83.33%



Support Vector Machine (SVM)

GridSearchCV · 10-fold cross-validation

Parameters:

kernel: linear/rbf/poly/sigmoid

C: [0.1, 1, 10]

CV

84.82%

Test

83.33%



Decision Tree ★ BEST

GridSearchCV · 10-fold cross-validation

Parameters:

criterion: gini/entropy

max_depth: 2-10 · min_samples: varied

CV

87.50%

Test

83.33%



K-Nearest Neighbors (KNN)

GridSearchCV · 10-fold cross-validation

Parameters:

n_neighbors: 1-10

algorithm: auto/ball_tree/kd_tree ·

p:[1,2]

CV

84.82%

Test

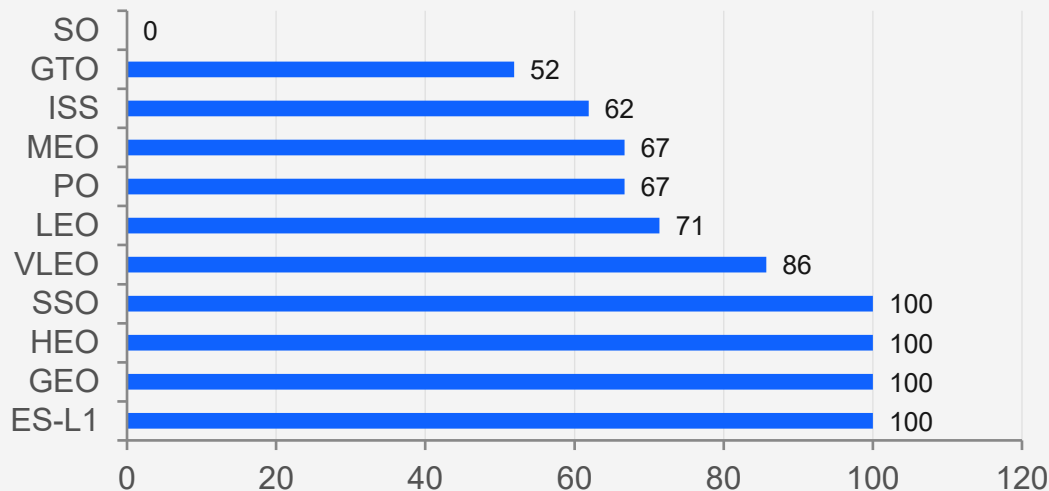
83.33%

EDA with Visualization — Results

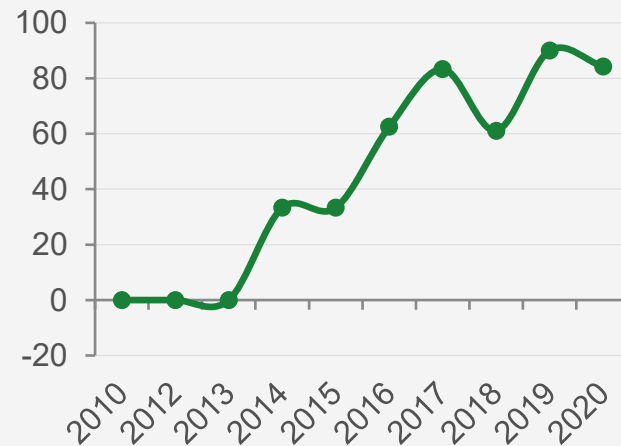
PATTERNS & TRENDS DISCOVERED

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Success Rate by Orbit Type (%)



Annual Success Trend (%)



- ES-L1, GEO, HEO, SSO orbits all achieved 100% success rate
- SO (Sub-Orbital) had 0% success — highest-risk mission profile
- Success grew from 0% (2010) to 90% (2019) — clear upward trajectory
- Higher flight numbers consistently correlate with improved landing success

EDA with SQL – Results

KEY QUERIES ON SPACEXTBL DATABASE

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Query 01

Unique Launch Sites

CCAFS SLC-40 · CCAFS LC-40
KSC LC-39A · VAFB SLC-4E

Query 02

Launches from KSC

22 total launches
from KSC LC-39A

Query 03

First RTLS Landing

Earliest True RTLS
date in dataset

Query 04

Failed ASDS in 2015

2 failures, both at
CCAFS SLC-40

Query 05

Drone Ship 4k–6k kg

Multiple F9 FT
boosters identified

Query 06

Top Outcome 2010–17

None None: 9 missions
True ASDS: 5 missions

Interactive Map Results — Folium

GEOSPATIAL ANALYSIS OF ALL LAUNCH SITES

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Launch Site	Latitude	Longitude	Location
CCAFS LC-40	28.5623°N	80.5774°W	Cape Canaveral, FL
CCAFS SLC-40	28.5632°N	80.5768°W	Cape Canaveral, FL
KSC LC-39A	28.5733°N	80.6469°W	Kennedy Space Center, FL
VAFB SLC-4E	34.6328°N	120.6107°W	Vandenberg, CA



Distance to Coastline — 0.93 km

CCAFS SLC-40 is just 0.93 km from the Atlantic coast — over-ocean trajectories for maximum safety



Success Markers — Green = Success

Majority green markers confirm dramatically improving booster recovery success over time



Early Failures — 2010–2013

Predominantly red markers in early years — no landing hardware (legs/grid fins) on v1.0 boosters



Coastal Sites — Safety by design

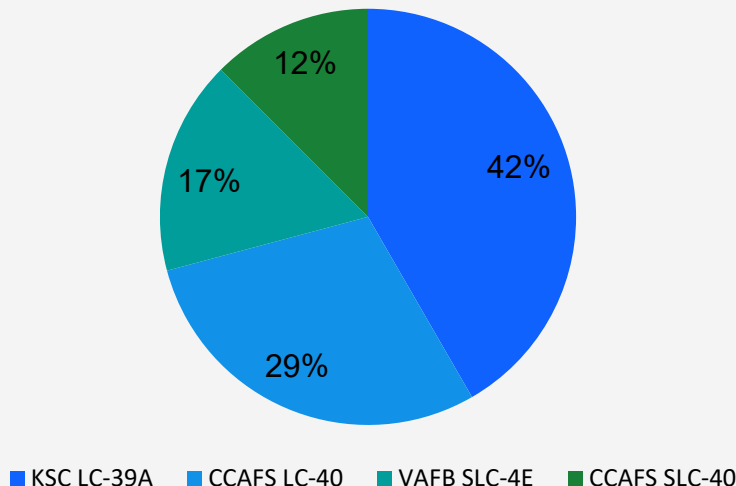
All US launch sites near coastlines — failed boosters safely land in the ocean away from population

Plotly Dash Dashboard — Results

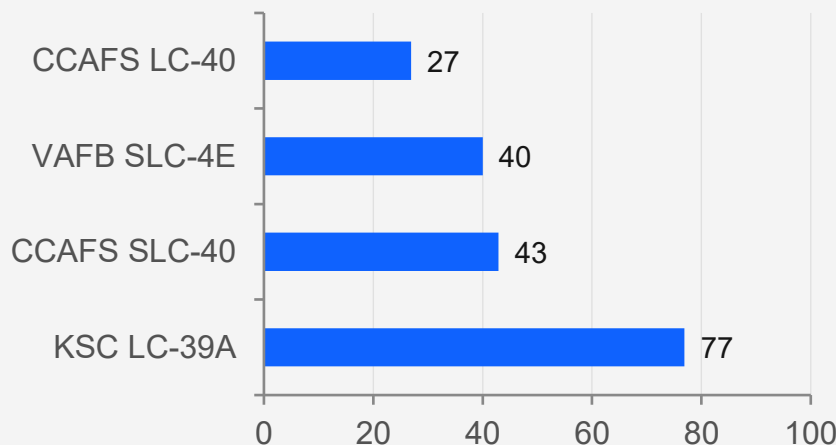
INTERACTIVE LAUNCH ANALYTICS

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Successful Launches by Site



Success Rate by Launch Site (%)



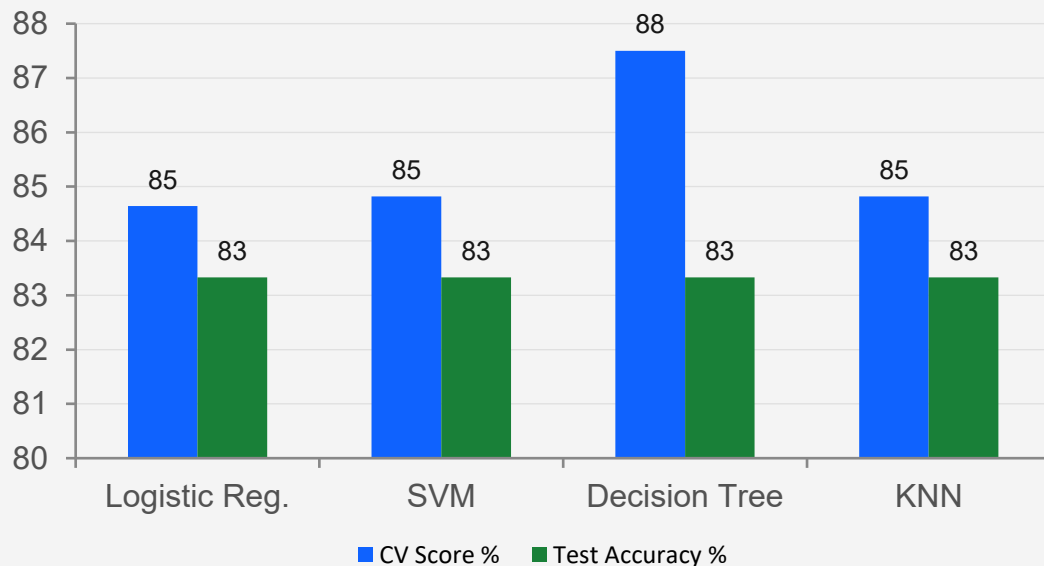
- 🚀 KSC LC-39A: Best site with 76.9% success rate and 10 successful launches (42% of all successes)
- 🚀 B5 booster version achieved 100% success rate across all tested payload mass ranges
- ⚠️ v1.0 booster: 0% success — early technology with no landing legs or grid fin hardware
- 📦 Optimal payload 2,000–5,500 kg shows the highest first-stage landing success probability

Predictive Analysis — Results

CLASSIFICATION MODEL PERFORMANCE COMPARISON

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CV Score vs Test Accuracy by Model



Logistic Regression

CV: 84.64% Test: 83.33%

SVM

CV: 84.82% Test: 83.33%

★ Decision Tree

CV: 87.50% Test: 83.33%

BEST CV

KNN

CV: 84.82% Test: 83.33%



Decision Tree — Best Model
CV Score: 87.50% · Test Accuracy: 83.33%

All 4 models tied at 83.33% test accuracy · Decision Tree leads with 87.5% CV score · Best params selected via GridSearchCV

Conclusion

SUMMARY OF FINDINGS & FUTURE DIRECTIONS

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Data Collection

Collected 90 Falcon 9 launches via SpaceX API + 364 historical records via Wikipedia BeautifulSoup scraping.



EDA Insights

KSC LC-39A best launch site (76.9%).
ES-L1/GEO/HEO/SSO orbits all at 100%.
GTO only 51.9% success.



Success Trend

Launch success improved 0% (2010) → 90% (2019), reflecting SpaceX's mastery of reusable rocketry.



ML Performance

Decision Tree achieved best CV score (87.5%). All 4 models tied at 83.3% test accuracy — strong consistency.



Business Value

Accurate predictions enable cost estimation: \$62M (reuse) vs \$165M (expendable) — 62% cost savings.



Future Work

Add 2021–2024 data, try XGBoost/Random Forest ensembles, integrate weather data & payload-type features.